DGLAP Equations

Quark Distribution

(1) $$\frac{\partial q_i(x,Q^2)}{\partial \ln(Q^2)} = \frac{\alpha_s(Q^2)}{2\pi}\left[\int_x^1 \frac{dy}{y} P_{qq}\left(\frac{x}{y}\right) q_i(y) + \int_x^1 \frac{dy}{y} P_{qg}\left(\frac{x}{y}\right) g(y)\right]$$

Gluon Distribution

(2) $$\frac{\partial g(x,Q^2)}{\partial \ln(Q^2)} = \frac{\alpha_s(Q^2)}{2\pi}\left[\int_x^1 \sum_{i=1}^{2n_f} \frac{dy}{y} P_{gq}\left(\frac{x}{y}\right) q_i^+(y) + \int_x^1 \frac{dy}{y} P_{gg}\left(\frac{x}{y}\right) g(y)\right]$$

where $q_i^+ = q_i + \bar{q}_i$

when summing over all $i = 1, \ldots, 2n_f$, eq (1) becomes:

(3) $$\frac{\partial \sum_{i=1}^{2n_f} q_i(x,Q^2)}{\partial \ln(Q^2)} = \frac{\alpha_s(Q^2)}{2\pi}\int_x^1 \frac{dy}{y}\left[\sum_{i=1}^{2n_f} P_{qq}\left(\frac{x}{y}\right) q_i(y) + 2n_f P_{qg}\left(\frac{x}{y}\right) g(y)\right]$$

then express (2) and (3) (using $\Sigma = \sum_{i=1}^{n_f} q_i + \bar{q}_i$) as:

$$\frac{\partial}{\partial \ln(Q^2)}\begin{bmatrix} \Sigma^{2n_f}(x,Q^2) \\ g(x,Q^2) \end{bmatrix} = \frac{\alpha_s(Q^2)}{2\pi}\int_x^1 \frac{dy}{y}\begin{bmatrix} P_{qq} & 2n_f P_{qg} \\ P_{gq} & P_{gg} \end{bmatrix}\left(\frac{x}{y}\right)\begin{bmatrix} \Sigma(y,Q^2) \\ g(y) \end{bmatrix}$$

One-loop Splitting Functions ($z = \frac{x}{y}$)

$$P_{qq}(z) = C_F\left[(1+z^2)\left(\frac{1}{1-z}\right)_+ + \frac{3}{2}\delta(1-z)\right]$$

$$P_{qg}(z) = T_F\left[(1-z)^2 + z^2\right]$$

$$P_{gq}(z) = C_F\,\frac{(1-z)^2+1}{z}$$

$$P_{gg}(z) = 2C_A\left[\frac{z}{(1-z)_+} + \frac{1-z}{z} + z(1-z)\right] + \left(\frac{11}{6}C_A - \frac{2}{3}T_F n_f\right)\delta(1-z)$$

where

$C_F = 4/3$ $\quad$ n_f = number of quark flavors

$T_F = 1/2$ $\quad$ $C_A = 3$

Expanded and Simplified Integrals (2x2 matrix DGLAP)

P_{qq}:

$$\int_x^1 \frac{dy}{y} P_{qq}(z)\, \Sigma(y) = \int_x^1 \frac{dy}{y} C_F \left[\frac{1+(z)^2}{(1-z)_+} + \frac{3}{2}\delta(1-z)\right] \Sigma(y)$$

$$= C_F \int_x^1 \frac{dy}{y} \frac{1+z^2}{1-z}\left[\Sigma(y) - 2z\,\Sigma(x)\right] + C_F\,\Sigma(x)\left[\frac{3}{2} + 2\ln(1-x)\right]$$

P_{qg}:

$$2n_f \int_x^1 \frac{dy}{y} P_{qg}(z)\, g(y) = 2n_f \int_x^1 \frac{dy}{y} T_F\left[(1-z)^2 + z^2\right] g(y)$$

P_{gq}:

$$\int_x^1 \frac{dy}{y} P_{gq}(z)\, \Sigma(y) = C_F \int_x^1 \frac{dy}{x}\left[1+(1-z)^2\right] \Sigma(y)$$

P_{gg}:

$$\int_x^1 \frac{dy}{y} P_{gg}(z)\, g(y) = 2C_A \int_x^1 \frac{dy}{y}\left[\frac{z}{(1-z)_+} + \frac{1-z}{z} + z(1-z) + \left(\frac{11}{6}C_A - \frac{2}{3}T_F n_f\right)\delta(1-z)\right] g(y)$$

$$= 2C_A \int_x^1 \frac{dy}{y}\left\{\frac{z\left[g(y)-g(x)\right]}{1-z} + \left[\frac{1-z}{z} + z(1-z)\right] g(y)\right\} + g(x)\left\{2C_A\left[x + \ln(1-x)\right] + \frac{11}{6}C_A - \frac{2}{3}T_F n_f\right\}$$